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Estimating access to safe water based on users' images: an application of machine learning

Silvan Bolt [1], Luis Arturo Estrada Grimaldo[1], Karin Gallandat [2], Valentina Boeva [3]
[1]ETH Zurich, D-INFK [2]Eawag [3]ETH AI Center, ETH Zurich

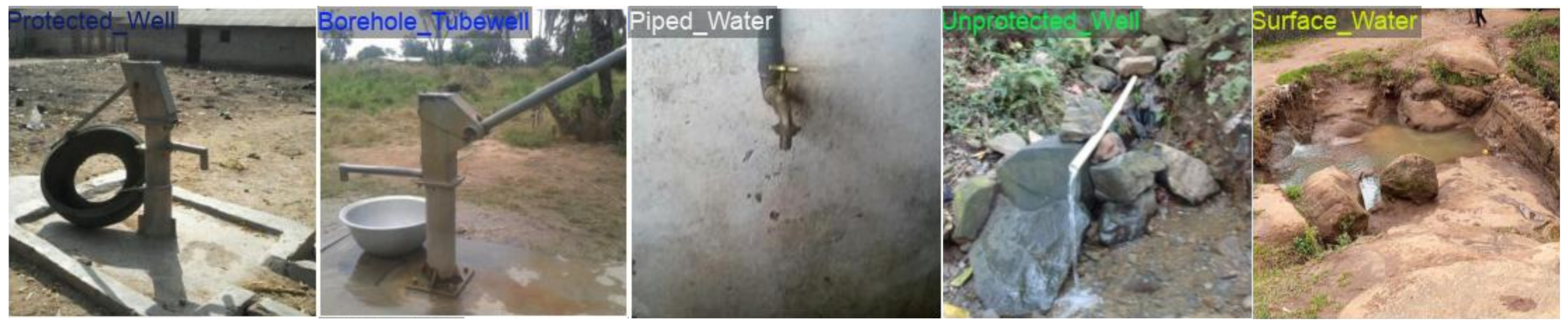

Figure 1: Water point samples. Left two images highlight visual similarity across distinct ground-truth categories.

1. Problem Statement

Access to safe drinking water is critical for public health, especially in regions affected by cholera and other waterborne diseases. However, granular information about the quality of water sources is often limited, and manually inspecting large collections of water point images is time-consuming.

We propose a computer vision system that predicts whether a water source image corresponds to an improved or unimproved water source, and further classifies the specific type of water infrastructure. This can help automate large-scale monitoring of water access and support progress toward SDG 6.1: universal access to safe drinking water.

2. Related work

Previous work in WASH research has focused on traditional ML algorithms to capture nonlinear relationships between socioeconomic, demographic, and geospatial variables in order to explain and predict access to safe water. Existing computer vision applications are more limited and have focused on satellite imagery (Dejito et al., 2021), water point detection from street-level images (Patel, 2024), or household water container analysis (Staddon et al., 2025). Our work complements this literature by classifying water source types directly from images.

3. Data

We use a dataset provided by the Water Point Data Exchange (WPdX), which includes metadata for nearly 200,000 images of water points from more than 25 countries. Each sample corresponds to one of the following categories:

- Improved: borehole/tubewell, piped water, protected spring, protected well, rainwater harvesting, sand/sub-surface dam.
- Unimproved: delivered water, surface water, unprotected well.

The data is highly imbalanced: improved water sources appear more frequently than unimproved water points.

4. Methods

We classify water point images into specific water source types and then map these predictions to the broader improved or unimproved categories. We selected YOLOv11m as the main model after preliminary experiments showed that it performed slightly better than a more complex Teacher–Student Distillation approach.

To reduce the effect of class imbalance, we used resampling strategies and used a custom loss function, assigning a larger cost to errors that confuse improved and unimproved sources.

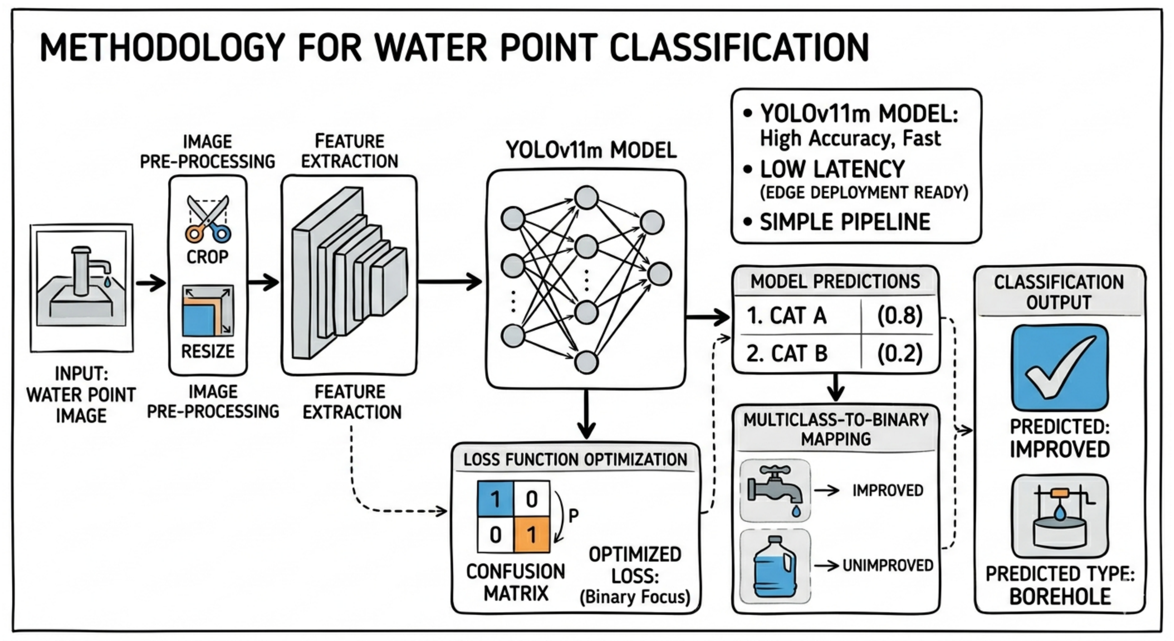

Figure 2: High-level overview of the YOLOv11m architecture, loss optimization, and final classification logic for water point assessment.

5. Results

The model achieved a top-1 accuracy of 73.87%, meaning that the most confident prediction matched the true water source type in nearly three quarters of the validation samples. Performance improved substantially when considering the model's ranked predictions: top-2 accuracy reached 94.65%, and top-3 accuracy reached 98.58%, showing the correct water source type is usually among the most likely alternatives. When the multiclass predictions were mapped back to the broader improved and unimproved categories, the model achieved a binary accuracy of 97.53%, indicating that most misclassifications occur between water source types within the same high-level group, rather than between improved and unimproved sources.

YOLO Multiclass Classification
Top-1: 73.87% | Binary: 97.53%

True Labels \ Predicted Labels	Borehole Tubewell	Piped Water	Protected Spring	Protected Well	Rainwater Harvesting	Sand or Sub-surface Dam	Delivered Water	Surface Water	Unprotected Well
Borehole Tubewell	16694	416	3791	1814	25	130	7	254	26
Piped Water	48	1472	16	89	9	91	0	119	13
Protected Spring	237	11	454	28	0	12	0	6	3
Protected Well	848	182	327	5041	9	54	0	194	75
Rainwater Harvesting	4	10	0	8	92	1	1	5	0
Sand or Sub-surface Dam	8	65	10	11	0	340	0	18	0
Delivered Water	1	1	0	0	0	0	3	2	0
Surface Water	13	25	13	41	1	11	0	413	17
Unprotected Well	0	9	4	14	0	11	0	14	1366

Figure 3: Confusion matrix for the multiclass classification task. The quadrants highlight the distinction between improved and unimproved sources.

6. Conclusions

The model is highly effective at identifying whether a water source belongs to the improved or unimproved category, while exact water source type classification remains more challenging due to visual similarity between infrastructure types, hidden underground infrastructure, and noisy ground-truth labels.

7. Future Work

- Improve training set quality by correcting wrong labels.
- Add epistemic uncertainty estimates so the system can flag low-confidence predictions.
- Enhance the web app to store user's location and system's prediction in a database.

References

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Contributions

Author contributions using the CRediT framework (Brand et al., 2015; Devos, 2024):
S1: Methodology, Software, Visualization, Investigation, Writing - Original Draft
S2: Methodology, Software, Visualization, Investigation, Writing - Original Draft
CG: Conceptualization, Supervision
AC: Supervision, Methodology, Project Administration